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END-OF-CHAPTER EXERCISES

Question 1.

A chord is 5 cm from the centre of a circle whose radius is 13 cm. Find the chord length.

Solution

Let AB be the chord and $OM \perp AB$.

Since the perpendicular from the centre to a chord bisects the chord,

$$AM = MB = \frac{AB}{2}.$$

Given,

$$OA = 13 \text{ cm}, \quad OM = 5 \text{ cm}.$$

In right triangle OMA ,

$$OA^2 = OM^2 + AM^2.$$

Hence,

$$13^2 = 5^2 + AM^2,$$

$$169 = 25 + AM^2,$$

$$AM^2 = 144,$$

$$AM = 12 \text{ cm}.$$

Therefore,

$$AB = 2AM = 24 \text{ cm}.$$

$AB = 24 \text{ cm}$

Question 2.

An arc subtends 70° at the centre. Find the angle subtended by the same arc at a point on the circle.

Solution

The angle subtended by an arc at the centre is twice the angle subtended by it at any point on the remaining circle.

Therefore,

$$\text{Angle at point on circle} = \frac{70^\circ}{2}.$$

Hence,

$$\boxed{35^\circ}.$$

Question 3.

The diameter of a circle is 26 cm. A chord of length 24 cm is drawn. Find its distance from the centre.

Solution

Radius,

$$r = \frac{26}{2} = 13 \text{ cm.}$$

Let $OM \perp AB$.

Then,

$$AM = MB = \frac{24}{2} = 12 \text{ cm.}$$

In right triangle OMA ,

$$13^2 = OM^2 + 12^2.$$

Therefore,

$$169 = OM^2 + 144,$$

$$OM^2 = 25,$$

$$\boxed{OM = 5 \text{ cm}}.$$

Question 4.

A circle has radius 15 cm. A chord is 9 cm from the centre. Find its length.

Solution

Let $OM = 9$ cm and $OA = 15$ cm.

Since OM bisects the chord,

$$AM = MB.$$

In right triangle OMA ,

$$15^2 = 9^2 + AM^2.$$

Thus,

$$225 = 81 + AM^2,$$

$$AM^2 = 144,$$

$$AM = 12 \text{ cm.}$$

Therefore,

$$AB = 2(12) = 24 \text{ cm.}$$

$$AB = 24 \text{ cm}$$

Question 5.

Prove that the perpendicular bisector of a chord passes through the centre of the circle.

Proof

Let AB be a chord of a circle with centre O , and let M be the midpoint of AB .

Thus,

$$AM = MB.$$

Also,

$$OA = OB$$

because they are radii.

OM is common to triangles OMA and OMB .

Therefore,

$$\triangle OMA \cong \triangle OMB$$

by SSS.

Hence,

$$\angle OMA = \angle OMB.$$

These two angles form a straight angle, so their sum is 180° .

Therefore,

$$\angle OMA = \angle OMB = 90^\circ.$$

Hence,

$$OM \perp AB.$$

Thus OM is the perpendicular bisector of AB and it passes through O .

The perpendicular bisector of every chord passes through the centre.

Question 6.

AB is a diameter and C is any point on the circle. Find $\angle ACB$.

Solution

Since AB is a diameter, the angle subtended by AB at any point on the circle is a right angle.

Therefore,

$$\angle ACB = 90^\circ.$$

Question 7.

$ABCD$ is cyclic. If $\angle A = 75^\circ$ and $\angle B = 110^\circ$, find $\angle C$ and $\angle D$.

Solution

Opposite angles of a cyclic quadrilateral are supplementary.

Therefore,

$$\angle A + \angle C = 180^\circ.$$

Hence,

$$75^\circ + \angle C = 180^\circ,$$

$$\angle C = 105^\circ.$$

Similarly,

$$\angle B + \angle D = 180^\circ.$$

Thus,

$$110^\circ + \angle D = 180^\circ,$$

$$\angle D = 70^\circ.$$

Question 8.

In cyclic quadrilateral $PQRS$,

$$\angle P = (2x + 10)^\circ, \quad \angle R = (3x - 20)^\circ.$$

Find x , $\angle P$ and $\angle R$.

Solution

Opposite angles of a cyclic quadrilateral add to 180° .

Therefore,

$$(2x + 10) + (3x - 20) = 180.$$

Thus,

$$5x - 10 = 180,$$

$$5x = 190,$$

$$\boxed{x = 38}.$$

Therefore,

$$\angle P = 2(38) + 10 = 86^\circ,$$

and

$$\angle R = 3(38) - 20 = 94^\circ.$$

$$\boxed{x = 38, \quad \angle P = 86^\circ, \quad \angle R = 94^\circ}$$

Question 9.

A chord is 16 cm long and its distance from the centre is 6 cm. Find the radius.

Solution

Let $AB = 16$ cm.

Then,

$$AM = MB = 8 \text{ cm.}$$

Let the radius be r .

In right triangle OMA ,

$$r^2 = 6^2 + 8^2.$$

Therefore,

$$r^2 = 36 + 64 = 100.$$

Hence,

$$\boxed{r = 10 \text{ cm}}.$$

Question 10.

A cyclic quadrilateral has side lengths 5, 5, 12, 12 units. Find its area.

Solution

Since the quadrilateral is cyclic, its opposite angles are supplementary.

With opposite equal sides, it can be divided into two congruent triangles having sides

$$5, \quad 12, \quad d$$

where d is the common diagonal.

Let the diagonal divide the quadrilateral into two triangles. For the corresponding configuration, the perpendicular height to the side 12 is

$$h = \sqrt{5^2 - \left(\frac{12}{2}\right)^2}$$

for the relevant isosceles triangle.

However, the two equal-side pairs form a rectangle-like cyclic configuration. The area is most conveniently obtained by Ptolemy's theorem.

Let the diagonals be d_1, d_2 . Since opposite sides are 5, 12,

$$d_1 d_2 = 5(12) + 5(12) = 120.$$

For a cyclic quadrilateral, Brahmagupta's formula gives

$$s = \frac{5 + 5 + 12 + 12}{2} = 17.$$

Therefore,

$$K = \sqrt{(17 - 5)(17 - 5)(17 - 12)(17 - 12)}.$$

Hence,

$$K = \sqrt{12 \cdot 12 \cdot 5 \cdot 5}$$

$$= \sqrt{3600}$$

$$\boxed{K = 60 \text{ square units}}.$$

Question 11.

For a cyclic quadrilateral, explain how to determine whether the circumcentre lies inside or outside it without drawing the circumcircle.

Solution

Take the perpendicular bisectors of any two sides of the quadrilateral.

Their intersection gives the circumcentre.

Now compare this point with the quadrilateral.

- If the centre lies inside the quadrilateral, the circumcentre is inside.
- If it lies outside the quadrilateral, the circumcentre is outside.

Another useful observation is that a chord is nearer to the centre when it is longer.

Thus, constructing perpendicular bisectors is the most direct method.

Intersection of perpendicular bisectors gives the circumcentre.

Question 12.

Two equal-length chords intersect. Prove that the corresponding two parts into which the chords are divided are equal.

Proof

Let equal chords AB and CD intersect at P .

Suppose

$$AB = CD.$$

By the intersecting-chords theorem,

$$PA \cdot PB = PC \cdot PD.$$

Also,

$$PA + PB = AB$$

and

$$PC + PD = CD.$$

Since

$$AB = CD,$$

we have

$$PA + PB = PC + PD.$$

Thus, the two pairs have equal sums and equal products.

Therefore, the corresponding members are equal.

Hence,

$$\boxed{PA = PC, \quad PB = PD}$$

(or the corresponding pairing after reversing the labels).

Question 13.

Construct a circle having a chord of length 6 cm at a perpendicular distance of 3 cm from the centre.

Construction

1. Draw $AB = 6$ cm.
2. Construct the midpoint M of AB .
3. Draw a perpendicular to AB through M .

4. On this perpendicular mark $OM = 3$ cm.
5. With centre O and radius OA , draw the circle.

Since

$$AM = 3 \text{ cm}, \quad OM = 3 \text{ cm},$$

by Pythagoras,

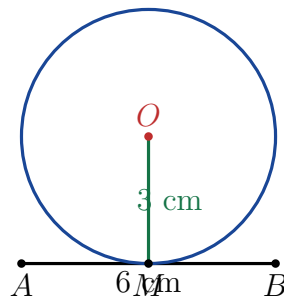
$$OA^2 = 3^2 + 3^2 = 18.$$

Thus,

$$OA = 3\sqrt{2} \text{ cm}.$$

Hence the required circle has radius

$$3\sqrt{2} \text{ cm}.$$



Question 14.

Prove that a rectangle is the only parallelogram that can be inscribed in a circle.

Proof

Let $ABCD$ be a parallelogram inscribed in a circle.

In a cyclic quadrilateral,

$$\angle A + \angle C = 180^\circ.$$

But in a parallelogram,

$$\angle A = \angle C.$$

Therefore,

$$2\angle A = 180^\circ,$$

so

$$\angle A = 90^\circ.$$

Similarly all four angles are right angles.

Therefore, the parallelogram is a rectangle.

Hence,

A parallelogram inscribed in a circle must be a rectangle.

Question 15.

Show that if a rectangle is inscribed in a circle, the intersection of its diagonals is the centre of the circle.

Proof

Let $ABCD$ be a rectangle inscribed in a circle.

The diagonals of a rectangle bisect each other.

Let them intersect at O .

Thus,

$$OA = OC$$

and

$$OB = OD.$$

Also, since all four vertices lie on the same circle,

$$OA = OB = OC = OD.$$

Therefore, O is equidistant from all four vertices.

Hence O is the centre of the circle.

The diagonals of an inscribed rectangle meet at the circumcentre.

Question 16.

Consider all chords of a circle having one fixed length. What shape is formed by the midpoints of all these chords?

Solution

All chords have the same length.

Equal chords of a circle are at equal perpendicular distance from the centre.

Therefore, the midpoint of every such chord is at the same distance from the centre.

Hence, the locus of all these midpoints is a circle whose centre is the centre of the original circle.

The midpoints form a circle concentric with the original circle.

Question 17.

In a circle with centre O , chords AB and AC are equal. Explain why O lies on the angle bisector of $\angle BAC$.

Proof

Given,

$$AB = AC.$$

Equal chords are equidistant from the centre.

Let the perpendiculars from O to AB and AC meet them at M and N .

Then,

$$OM = ON.$$

Also,

$$OA = OA.$$

Thus the right triangles OMA and ONA are congruent by RHS.

Therefore,

$$\angle MAO = \angle OAN.$$

Hence AO bisects the angle between AB and AC .

Therefore,

$$\boxed{AO \text{ is the angle bisector of } \angle BAC}.$$

Question 18.

Two parallel chords of lengths 10 cm and 24 cm lie on the same side of the centre. Their separation is 7 cm. Find the radius.

Solution

Let the distances of the two chords from the centre be x and $x + 7$.

For the 10 cm chord,

$$\frac{10}{2} = 5.$$

Thus,

$$r^2 = x^2 + 5^2.$$

For the 24 cm chord,

$$\frac{24}{2} = 12.$$

Since the longer chord is nearer to the centre,

$$r^2 = (x + 7)^2 + 12^2$$

would correspond to the farther chord, so instead take

$$r^2 = x^2 + 12^2$$

for the 24 cm chord and

$$r^2 = (x + 7)^2 + 5^2$$

for the 10 cm chord.

Therefore,

$$x^2 + 144 = (x + 7)^2 + 25.$$

Expanding,

$$x^2 + 144 = x^2 + 14x + 49 + 25.$$

Hence,

$$144 = 14x + 74,$$

$$14x = 70,$$

$$x = 5.$$

Therefore,

$$r^2 = 5^2 + 12^2$$

$$= 25 + 144 = 169.$$

Thus,

$$\boxed{r = 13 \text{ cm}}.$$

Question 19.

A regular hexagon is inscribed in a circle of radius r . Find its side length and the perpendicular distance of each side from the centre.

Solution

Join the centre O to two adjacent vertices A and B .

A regular hexagon divides the circle into six equal sectors.

Therefore,

$$\angle AOB = \frac{360^\circ}{6} = 60^\circ.$$

Since

$$OA = OB = r,$$

triangle AOB is equilateral.

Therefore,

$$AB = r.$$

Hence the side of the hexagon is

$$\boxed{r}.$$

Now let M be the midpoint of AB .

In right triangle OAM ,

$$OA = r, \quad AM = \frac{r}{2}.$$

Therefore,

$$\begin{aligned} OM^2 &= r^2 - \left(\frac{r}{2}\right)^2 \\ &= r^2 - \frac{r^2}{4} \\ &= \frac{3r^2}{4}. \end{aligned}$$

Hence,

$$\boxed{OM = \frac{\sqrt{3}}{2}r}.$$

Thus,

$$\boxed{\text{Side length} = r}$$

and

$$\boxed{\text{Distance of each side from centre} = \frac{\sqrt{3}}{2}r}.$$

Question 20.

$MNOP$ is cyclic and MN is a diameter. Determine $\angle MOP$ and $\angle MNP$.

Solution

Since MN is a diameter, the angle subtended by it at any point on the circle is 90° .

Therefore,

$$\boxed{\angle MPN = 90^\circ}.$$

However, $\angle MOP$ is a central angle and is not fixed without additional information about the position of P .

Thus,

$$\boxed{\angle MOP \text{ cannot be uniquely determined from the given information.}}$$

Question 21.

For cyclic quadrilateral $ABCD$, show that the exterior angle at a vertex equals the interior opposite angle.

Proof

Let CD be extended to E .

The exterior angle is

$$\angle CDE.$$

Since $ABCD$ is cyclic,

$$\angle DAB + \angle DCB = 180^\circ.$$

Also, the exterior angle and interior angle at D form a linear pair:

$$\angle CDE + \angle CDA = 180^\circ.$$

Therefore,

$$\angle CDE = 180^\circ - \angle CDA.$$

But in the cyclic quadrilateral,

$$\angle CDA + \angle CBA = 180^\circ.$$

Hence,

$$\angle CDE = \angle CBA.$$

Thus,

$$\boxed{\text{Exterior angle} = \text{interior opposite angle}}.$$

Question 22.

Justify the statement that no chord of a circle can be longer than its diameter.

Proof

Let AB be any chord of a circle with radius r .

Let M be the midpoint of AB and $OM = d$.

Then,

$$AB = 2\sqrt{r^2 - d^2}.$$

Since

$$d^2 \geq 0,$$

we have

$$r^2 - d^2 \leq r^2.$$

Therefore,

$$AB \leq 2r.$$

But $2r$ is the diameter.

Hence,

Every chord is at most as long as the diameter.

Equality occurs only when

$$d = 0,$$

i.e. when the chord passes through the centre and is therefore a diameter.

Question 23.

A point A lies inside a circle with centre O . Show that the shortest chord passing through A is perpendicular to OA .

Proof

Consider any chord BC passing through A .

Let $OM \perp BC$.

Since the perpendicular from the centre to a chord bisects the chord,

$$BM = MC.$$

If OA is not perpendicular to BC , then $A \neq M$.

Since M is the midpoint of BC ,

$$BC = 2BM.$$

The chord is shortest when its midpoint is as close as possible to the centre.

Among all lines through A , the shortest perpendicular distance from O to the line occurs when the line is perpendicular to OA .

Thus, when

$$BC \perp OA,$$

the midpoint of BC lies on OA and gives the minimum chord length.

Therefore,

The shortest chord through A is perpendicular to OA .

Question 24.

Use the given figure to explain why an angle subtended by a diameter in a semicircle is 90° .

Solution

Let AB be a diameter and C be any point on the semicircle.

Join OC .

Since

$$OA = OC$$

and

$$OB = OC,$$

triangles OAC and OBC are isosceles.

Let

$$\angle OCA = \angle OAC = x$$

and

$$\angle OCB = \angle OBC = y.$$

Since A, O, B are collinear,

$$\angle AOC + \angle COB = 180^\circ.$$

Using the angle sum of the two isosceles triangles,

$$2x + \angle AOC = 180^\circ$$

and

$$2y + \angle COB = 180^\circ.$$

Adding appropriately gives

$$x + y = 90^\circ.$$

But

$$\angle ACB = x + y.$$

Therefore,

$$\boxed{\angle ACB = 90^\circ}.$$

Question 25.

Two chords CC' and DD' are perpendicular to a diameter AB . If M and M' are their midpoints, prove that $MM' \perp AB$.

Proof

Let O be the centre of the circle.

Since CC' and DD' are perpendicular to the diameter AB , the perpendicular from the centre to each chord bisects it.

Therefore,

$$MC = MC'$$

and

$$M'D = M'D'.$$

More importantly, O, M, M' all lie on the perpendicular bisectors of the corresponding chords.

Hence both M and M' lie on the same line perpendicular to AB .

Therefore,

$$\boxed{MM' \perp AB}.$$

Question 26.

Use the given figure to justify that the sum of the opposite angles of a cyclic quadrilateral is 180° .

Solution

Let $ABCD$ be a cyclic quadrilateral with centre O .

Join OA, OB, OC, OD .

The angles at the centre corresponding to the arcs satisfy

$$\angle AOB + \angle BOC + \angle COD + \angle DOA = 360^\circ.$$

The inscribed angle theorem says that an angle at the circumference is half the central angle subtending the same arc.

Thus,

$$\angle A + \angle C = \frac{1}{2} (\angle BOC + \angle DOA) .$$

Similarly,

$$\angle B + \angle D = \frac{1}{2} (\angle AOB + \angle COD) .$$

Adding the two opposite-angle sums,

$$\angle A + \angle C + \angle B + \angle D = \frac{1}{2}(360^\circ) = 180^\circ + 180^\circ .$$

Therefore,

$$\boxed{\angle A + \angle C = 180^\circ}$$

and

$$\boxed{\angle B + \angle D = 180^\circ} .$$

Hence, the opposite angles of a cyclic quadrilateral are supplementary.

CHAPTER 5 – QUICK REVISION

Perpendicular from centre to chord bisects the chord

Equal chords are equidistant from centre

Longer chord is nearer to the centre

Angle at centre = $2 \times$ angle at circumference

Angle in a semicircle = 90°

Opposite angles of cyclic quadrilateral = 180°

Chord length = $2\sqrt{r^2 - d^2}$

Diameter is the longest chord